



Nursing Practice Guideline

Percutaneous Biliary Drainage Catheters

Reference #: NPG 1

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Percutaneous Biliary Drainage Catheters. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

Table of Contents

Introduction	1
Risk Statement	1
Principles	2
1.0 General Information and Care	2
2.0 Catheter Site Dressings and Drainage System Change	4
3.0 Flushing of Catheter	6
4.0 To Obtain a Specimen of Bile	7
5.0 Removal of Biliary Drainage Catheter.....	7
6.0 Discharge Planning	
Related Documents	10
References	11
Authorisation & Version Control.....	11

Principles

Observe the five moments for hand hygiene and aseptic technique handling or manipulating biliary drainage catheters. Follow the risk assessment outlined in the SCGOPHCG Infection Prevention and Control Manual - Aseptic Technique Policy. This must include use of appropriate personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.

1.0 General Information and Care

1.1 Medical Requirements

Informed written consent as per North Metropolitan Health Service Consent to Treatment Policy, and in compliance with the WA Health Consent to Treatment Policy (OD0657/16 v 2.0).

Ensure peripheral vascular catheter or power port™ CVC / PICC line insitu.

Prophylactic antibiotic prescribed for cases of obstructive jaundice^{1,2,3}

Coagulation profile, FBC, LFTs and U & Es within 24 hours of procedure and consider correction of coagulopathy.^{1, 2, 3.}

Review procedural instructions for specific post insertion orders.

1.2 Safety Information

Observe for procedure-related complications: bleeding including melaena, pain, bile leak, cholangitis, sepsis, bowel perforation, acute pancreatitis, biloma, intrahepatic haematoma, pneumothorax, and pleural effusion (transhepatic route)^{1,2,3,4}. Report any signs of blood in drainage fluid to medical officer.

Catheters used for percutaneous biliary drainage are predominantly pigtail catheters with a locking mechanism e.g. Merit medical resolve™ or Cook mac lock™ and should be removed as per manufacturer's instructions (see section 5).

1.3 Nursing Management

General Information

Percutaneous transhepatic cholangiography (PTC): is a diagnostic procedure for imaging the biliary system, it is mostly used as the first step in percutaneous transhepatic biliary drainage.

Percutaneous transhepatic biliary drainage (PTBD): is used to treat blockage or narrowing of the bile duct resulting from strictures, tumours, cirrhosis, inflammation or infection^{1,3} A catheter is placed under radiological guidance by the transhepatic route into the common bile duct.

Biliary drainage may be achieved with an:^{2,3}

1. External biliary catheter– from the outside of the body to the bile duct
2. External-internal biliary catheter –from the outside of the body to the bile duct and extending into the duodenum, allows bile to drain past an area of obstruction internally and to a drainage bag which may then be spigotted to allow all bile to drain internally only.

Percutaneous transhepatic biliary stenting (PTBS) An internal biliary stent – (metal or mesh) may be placed from the bile duct to the duodenum this may be inserted initially or when either External biliary catheter or External-internal biliary catheter has been in place.

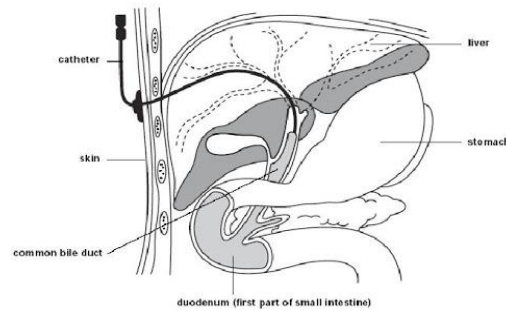


Figure 1⁶

Percutaneous Cholecystostomy catheter / tube

Cholecystostomy catheters are placed as an alternative to cholecystectomy for acute cholecystitis, where surgery is contraindicated due to critical illness, age or comorbidities. Percutaneous cholecystostomy allows immediate decompression of an acutely inflamed gall bladder reducing symptoms and acute inflammatory response. It is usually a temporary intervention.

- Placed under radiological guidance by the transhepatic or transperitoneal route directly into the gallbladder lumen

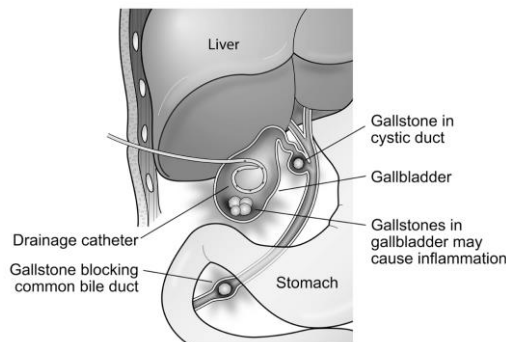


Figure 2⁷

Biliary drainage can vary in colour and consistency; it may be thin to thick, golden yellow, brown, green or occasionally clear to white fluid.

Pre Insertion

- Patient must be fasted; no food for a minimum of 6 hours and no fluids for a minimum of 2 hours prior to the procedure.
- A Radiology checklist must be completed for all patients having an invasive procedure.
- Refer to Nursing Practice Guideline 26: Radiological Examinations for pre-procedure preparation and care.
- Administer prescribed antibiotics preferably within one hour of procedure.^{1,3}

Post Insertion

- Refer to post procedure instructions.
- Post procedure core physiological observation document using the observation and response chart (ORC).
- The catheter should be connected to a 3 way tap then to connecting tubing (CT) which has a 2 way on / off tap. The end of the CT may be spigotted or

connected to a drainage bag. If connected to a drainage bag check connections, patency and that there are no kinks in the tubing.

If a 3 way tap is not connected – attach between the catheter and the vinyl connecting tubing using an aseptic non-touch technique. Place needleless bung on one port.

- Anchor the drainage bag tubing to the skin with adhesive tape (Fixomull®) or a catheter fixation device (e.g., Flexi Trak) to prevent excessive traction.
- If free drainage is ordered ensure the 3-way tap is in the open position and drainage bag with a catheter tip connection is attached.
- The catheter will be secured at the insertion site with a fixation device e.g. Statlock™ or Griplock™. A Statlock™ must be the correct size for the gauge of catheter; a Griplock is one size for all catheters.²
- A Statlock™ will provide a more secure fixation than a Griplock™ and should be considered if accidental removal or longer-term use is anticipated e.g., restless patient, patients being discharged home with a catheter in situ.
- A Statlock can be uncomfortable for the patient to lie on and may cause pressure on the skin, particularly if multiple catheters are in situ. Inspect the skin frequently for injury and educate the patient to avoid lying on the drain(s).
- Remove and replace primary fixation device Statlock™ or Grip-Lok™ every 7 days and PRN if soiled, or lifting off the skin.
- Change secondary occlusive dressing as required. A small absorbent pad may be required under the occlusive dressing, if there is any bile leak around the catheter, to protect surrounding skin. Dressing may then need to be changed more frequently.
- Securing the catheter with a suture may also be considered.
- Empty drainage bag and record the volume of bile drained each shift: record on fluid balance chart. Total 24-hour drainage volume must be recorded on Fluid Balance Summary. Document colour and description of drainage in inpatient record.
- If catheter is not draining it may be occluded or displaced requiring replacement, report to medical officer.
- Observe the insertion site for leakage, inflammation or discharge and report to Medical Officer. Report any pain at insertion site.

2.0 Catheter Site Dressings and Drainage System Change

2.1 Nursing Management

The secondary occlusive dressing, fixation device and drainage system should be changed every 7 days or more frequently if indicated

2.2 Procedural Information

Requirements

- Fixation device (universal Grip-Lok™ or StatLock™ of appropriate size)
- Large transparent occlusive dressing (Tegaderm™)
- Normal saline 0.9%

- Dressing pack
- Adhesive tape (e.g., Fixomull™.)
- Wet strength bag
- Sterile gloves
- 1 x Skin barrier wipe
- 1 x Adhesive remover wipe
- Personal Protective Equipment (PPE)

Additional requirements when drainage system change required.

- Sterile drainage bag
- Connecting Tubing (CT) - male luer lock to catheter tip fitting
- 3-way tap (only if flushing required)
- Needleless bung (only if flushing required)
- Large 70% alcohol wipe

Procedure

1. Perform hand hygiene.
2. Clean trolley with pre-diluted detergent, or a detergent wipe. Allow to air dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare field.
5. Open dressing pack using corners and add equipment to sterile field using non-touch technique
6. Explain procedure to patient and position patient for ease of access to catheter, placing an absorbent sheet underneath.
7. Perform hand hygiene and don non-sterile gloves.
8. Ensure catheter is secured to patients' skin with anchor tape / fixation device.
9. Remove secondary dressing minimise catheter manipulation during removal by using adhesive remover wipe.
10. Statlock™ Removal Technique:
 - Disengage: open StatLock™ device retainer by pressing the tab and lifting the lid
 - Carefully lift the catheter from the retainer
 - Dissolve: lift the edge of the anchor pad using adhesive remove wipe and continue to remove fully
11. Perform hand hygiene, don sterile gloves.
12. If drainage system change is required, attach drainage bag and 3 way tap to CT.
13. Place sterile towel beneath catheter.
14. Clean around entry site with normal saline 0.9%.
15. Inspect surrounding skin for redness, tenderness, swelling or irritation, if infection suspected swab site for culture and inform medical officer.
16. StatLock™ application technique:
 - Apply skin prep wipe allow to dry.
 - Secure catheter to the StatLock™ before placing pad on the skin, bring StatLock™ pad into position. Stabilise catheter between fingers and stretch to fit into retainer.
 - Close the retainer lid.
 - Peel away the StatLock™ backing one side at a time, place on skin.
17. Ensure catheter tubing is not kinked/ twisted.
18. Cover the fixation device with a transparent occlusive dressing ensuring a good seal.
19. If drainage bag change is required, clean connection between catheter and 3-way tap with large 70% alcohol wipe and allow to dry.

20. Disconnect catheter from 3 way tap and attach new drainage system.
21. Ensure the catheter is secured below the hipline with adhesive tape e.g., Fixomull™
22. Dispose of waste.
23. Remove PPE and perform hand hygiene.
24. Clean trolley with pre-diluted detergent, or a detergent wipe.
25. Perform hand hygiene.
26. Document and record date of change in patient's notes and Nursing Care Plan

3.0 Flushing of Catheter

3.1 Medical Requirements

Flushing of a catheter is only to be performed on request from the Medical Officer and is to be prescribed on the WA Hospital Medication Chart (MR 805)). Instructions must include the volume of the flush and if the flush is to be aspirated or left in situ to drain.

3.2 Safety Information

If any excess pressure is required to flush, or if the patient experiences pain, stop the procedure and notify the Medical Officer.

All injectable medicines drawn up in a syringe should be labelled **immediately**. Labelling is not required when the preparation and bolus administration of a single medicine is one continuous process, and where the syringe **does not** leave the hands of the person who prepared it and that same person administers the medicine.

3.3 Nursing Management

Post flushing procedure the catheter must have a drainage bag attached.

3.4 Procedural Information

Requirements

- Personal protective equipment
- Non-sterile gloves
- Non-sterile kidney dish / tray
- 10ml Luer lock syringe(s)
- Drawing up needle (s)
- 0.9% Sodium Chloride 10ml injection ampoule(s)
- Large 70% alcohol wipe
- Absorbent sheet

Procedure

1. Perform hand hygiene
2. Assemble equipment in non-sterile kidney dish / tray and check patient identity against prescription order in WA Hospital Medication Chart
3. Position the patient for ease of access to the catheter placing an absorbent sheet under the catheter.
4. Perform hand hygiene and don non-sterile gloves.
5. Protect key parts using a non-touch technique. Scrub the needleless bung on the outlet port of the 3-way tap with large 70% alcohol wipe for 30 seconds. Allow to air dry for 30 seconds.
6. Turn the 3-way tap off to the drainage bag and connect syringe with Sodium Chloride 0.9% flush to needleless bung.

7. Aspirate gently to commence flow and gently inject prescribed volume of Sodium Chloride 0.9% into catheter, do not use force.
8. Either aspirate gently to withdraw the same amount of fluid (if unable to withdraw fluid inform medical officer) OR leave fluid in if ordered.
9. Turn the 3-way tap - off to the needless bung, and disconnect the syringe.
10. Ensure 3-way tap is left in open position to the drainage bag.
11. Check the system for patency and kinking.
12. Dispose of waste.
13. Remove PPE and perform hand hygiene.
14. Clean trolley with pre-diluted detergent, or a detergent wipe.
15. Perform hand hygiene.
16. Record completion of Sodium Chloride 0.9% flush on WA Hospital Medication Chart (MR805); if any difficulties are encountered inform medical officer, shift coordinator and record in inpatient record.

4.0 To Obtain a Specimen of Bile

4.1 Medical Requirements

Specimen is obtained at the request of the Medical Officer complete request form or Computerised Provider Order Entry (CPOE)

4.2 Procedural Information

Requirements

- Non-sterile kidney dish / tray
- Luer lock 10 ml syringe.
- Personal Protective Equipment (PPE)
- Sterile yellow top specimen container
- Large 70% alcohol wipe

Procedure

1. Perform hand hygiene
2. Gather equipment in non-sterile kidney dish / tray
3. Perform hand hygiene
4. Don non-sterile gloves
5. Protect key parts using a non-touch technique. Scrub the needless bung on 3-way tap with a large 70% alcohol wipe for 30 seconds and allow to air dry for 30 seconds.
6. Connect syringe to needless bung with 3-way tap off to the drainage bag.
7. Gently aspirate to withdraw 1-10 ml of bile, disconnect the syringe and transfer bile into yellow top jar.
8. Turn the 3-way tap open to patient and drainage bag.
9. Check the system for patency and kinking.
10. Dispose of waste appropriately.
11. Remove gloves and perform hand hygiene.
12. Complete collection details on request form and clearly label specimen container for transport.
13. Document in the inpatient record.

5.0 Removal of Biliary Drainage Catheter

5.1 Medical Requirements

Removal of the biliary drainage catheter is at the direction of the medical team and must be documented in the inpatient record.

5.2 Safety Information

Catheters used for percutaneous biliary drainage are predominantly pigtail catheters with a locking mechanism e.g. Merit Medical Resolve™ or Cook Mac Lock™ and should be removed as per manufacturer's instructions. Unlocking for removal is dependent on catheter type. Do not cut pigtail catheter unless specified by the manufacturer (instructions should be sent from radiology post insertion).

Failure to uncoil a pigtail catheter prior to removal can cause severe pain and internal tissue damage. It can also cause breakage of the drain at the distal tip and retention of drain fragments. ³

If there is undue resistance or discomfort removing the catheter, **stop** the procedure, **tape** the catheter to skin to prevent it retracting into the wound and apply dressing. Inform the medical officer.

Post removal, check the catheter tip is intact, and document in patient's notes.

5.3 Nursing Management

Ensure adequate analgesia (if required) is administered prior to removal of biliary drainage catheter.

5.4 Procedural Information

Requirements

- Normal saline 0.9%
- Dressing pack
- 1 x adhesive remover wipe
- Sterile stitch cutter, if required
- Sterile dressing (e.g. Mepilex border)
- Wet strength bag
- Protective Equipment (PPE)
- Forceps / blunt scissors (Mac-Loc catheter)

Pigtail catheter loop unlocking device (Merit Medical catheter)

Procedure

1. Perform hand hygiene.
2. Clean trolley with pre-diluted detergent, or a detergent wipe. Allow to dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare field.
5. Open dressing pack and add equipment to sterile field using non-touch technique.
6. Perform hand hygiene and don PPE (non-sterile gloves) and eyewear
7. Remove fixation device using adhesive solvent.
8. Remove PPE and perform hand hygiene.
9. Clean around the insertion site with normal saline 0.9%.
10. Remove suture if present.
11. Apply skin prep to surrounding skin
12. Unlock catheter if required (as per manufacturer's instructions: see below)
13. Unlocking Pigtail Catheters:

A. Cook medical: Mac-Loc™ locking mechanism

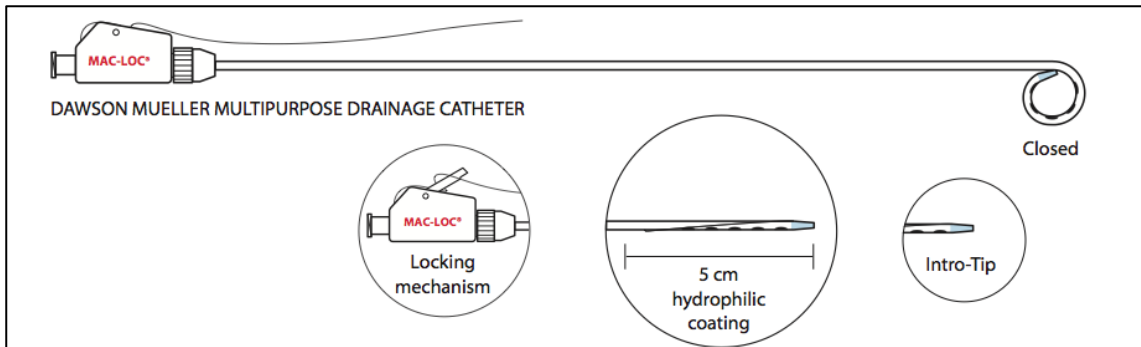


Figure 3 Locked position

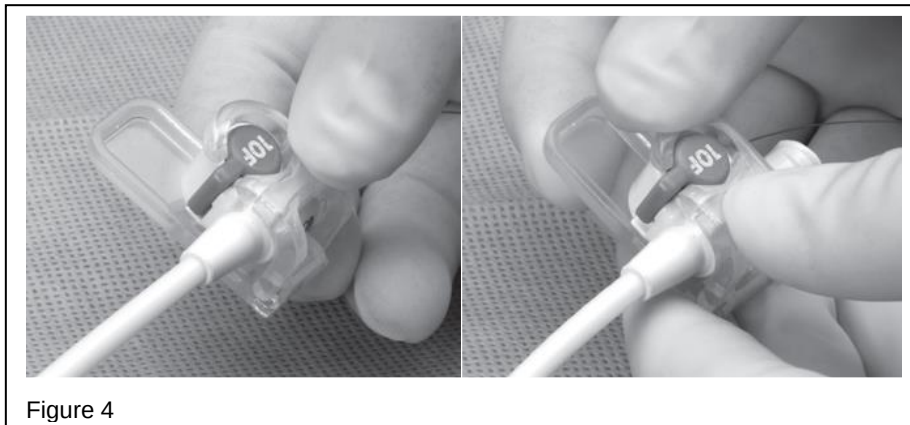
To unlock

- Whilst stabilising the Mac-Loc catheter hub assembly with one hand, position a small blunt object (approximately the size and shape of a ball point pen or small forceps) into the Mac-Loc release notch.
- Pry forward until the locking cam lever is free.

B. Merit Medical Resolve Catheter

OPTION 1: Unlocking with repositioning tool / key

- Align the opening of the round section of the repositioning tool in line with the handle of the suture locking mechanism
- Bring the flat back of the repositioning tool in line with the handle of the suture locking mechanism.
- Bring the flat back of the repositioning tool around the catheter.
- Gently squeeze together; this disengages the catheter locking mechanism.
- Rotate handle clockwise. Hub is now unlocked.



OPTION 2:

Cut the hub off the drainage catheter and sever the suture. This will release the suture and the pigtail loop

Caution: The suture will no longer be secured to the catheter, take care to remove both the suture and the catheter.

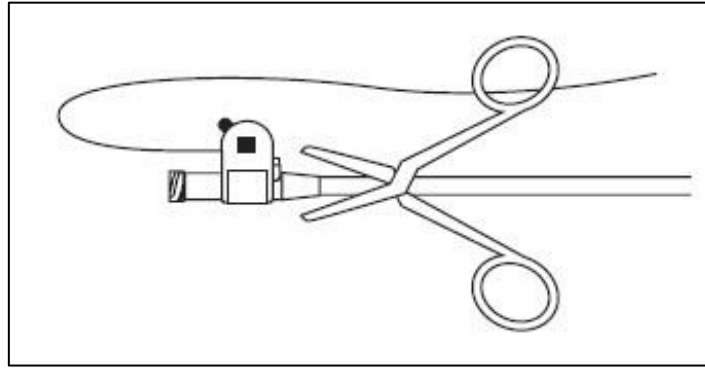


Figure 5

14. Don PPE (non-sterile gloves).
15. Remove the drain in a continuous motion by exerting gentle traction on the tube, while supporting skin surrounding the drain, ensure catheter and string intact.
16. Apply occlusive dressing to insertion site.
17. Dispose of waste.
18. Remove PPE and perform hand hygiene.
19. Clean trolley with pre-diluted detergent, or a detergent wipe. Allow to air dry.
20. Perform hand hygiene.
21. Document removal of drain in the patient notes and record amount of drainage on Fluid Chart

6.0 Discharge of Patient with PTBD Catheter. ⁵

6.1 Safety Information

Ensure competence of patient / family / carers or home service providers with care of catheter.

6.2 Medical Requirements

Flushing of a catheter is only to be performed on request from the Medical Officer and is to be prescribed on appropriate community services documentation. Instructions must include the volume and frequency of the flush and if the flush is to be aspirated or left in situ to drain.

6.3 Nursing Management

Patient / carer education should include:

- Infection prevention including hand hygiene.
- Keeping catheter dressing intact and dry, use of shower covers
- Emptying drainage bag and recording volume of drainage.
- Avoid traction on catheter, prevent from kinking or twisting.
- Ensure catheter is anchored to the skin.
- Observe for signs of infection, fevers, pain, redness, tenderness at insertion site.
- Observe for bile leakage / bleeding around insertion site.

Refer to appropriate community nursing service e.g., Home Link or Silver Chain post-acute care team.

Provide contact details for Residential Care Line if patient is discharged to a residential aged care facility.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010

- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010

Work Health and Safety Act 2020Policy:

- WA Health Mandatory Policy Consent to Treatment Policy
- WA Department of Health Mandatory Policy 0131/20 High Risk Medication Policy.
- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Control manual -Aseptic Technique Policy.
- SCGH Patient Identification and Procedure Matching Policy
- SCGH Hospital Wide Policy Communicating for Safety

Guidelines

- SCGOPHCG Nursing Practice Guideline 65 Wound Drain Management
- SCGOPHCG Nursing Practice Guideline 26 Radiological Examinations

References

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Authorisation & Version Control

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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